

1. Create a Jenkins project that automates the execution of a simple Java program stored in a GitHub repository. The project should:
 - a. Clone the GitHub repository containing the Java program.
 - b. Poll the repository every 5 minutes for changes.
 - c. Compile and run the Java program using "Execute Windows batch command" in Jenkins.
 - d. Display the program output in the Jenkins Console Output.

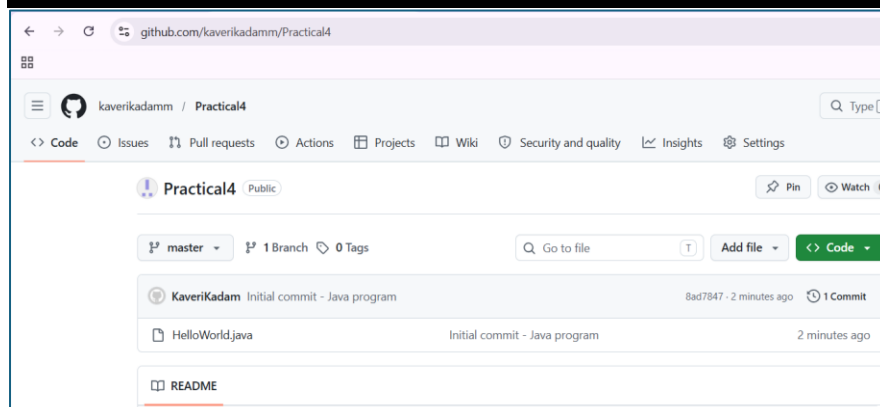
Create a java Program

```
public class HelloWorld{  
    public static void main(String[] args){  
        System.out.println("Hello World!");  
    }  
}
```

Create a public repository Practical4 on github.

Initialize Git and Push.

```
kaver@KAVERI MINGW64 ~  
$ cd D:  
  
kaver@KAVERI MINGW64 /d  
$ cd D:/MCA/"SEM II"/"DevOps Lab"/Pract4  
  
kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Pract4  
$ git init  
Initialized empty Git repository in D:/MCA/SEM II/DevOps Lab/Pract4/.git/  
  
kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Pract4 (master)  
$ git add .  
  
kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Pract4 (master)  
$ git commit -m "Initial commit - Java program"  
[master (root-commit) 8ad7847] Initial commit - Java program  
1 file changed, 5 insertions(+)  
create mode 100644 HelloWorld.java  
  
kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Pract4 (master)  
$ git remote add origin https://github.com/kaverikadamm/Practical4.git  
  
kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Pract4 (master)  
$ git push -u origin master  
Enumerating objects: 3, done.  
Counting objects: 100% (3/3), done.  
Delta compression using up to 12 threads  
Compressing objects: 100% (2/2), done.  
Writing objects: 100% (3/3), 323 bytes | 323.00 KiB/s, done.  
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)  
To https://github.com/kaverikadamm/Practical4.git  
* [new branch]      master -> master  
branch 'master' set up to track 'origin/master'.
```



Start tomcat: localhost:8080 and then start Jenkins localhost:8090.

Create a pipeline as java_pipeline.

Script:

```
pipeline {
  agent any

  environment {
    BUILD_VERSION="1.0"
  }

  tools{
    jdk 'java17Home'
  }

  triggers {
    pollSCM('H/5 * * * *')
  }

  stages {
    stage('Clone Repository') {
      steps {
        script {
          def workspace = env.WORKSPACE
          def cloneJava= "${workspace}\\cloneJava"

          // Delete old repo if exists
          if (fileExists(cloneJava)) {
            bat "rmdir /S /Q \"${cloneJava}\""
            echo "Removed existing Repo folder."
          }

          bat "mkdir cloneJava"
          bat "git clone https://github.com/kaverikadamm/Practical4.git \"${workspace}\\cloneJava\""
        }
      }
    }

    stage('Compile Java Program') {
      steps {
        script{
          bat """
          echo Compiling Java Program
          cd cloneJava
          javac HelloWorld.java
          """
        }
      }
    }
  }
}
```

```

stage('Run Java Program'){
    steps{
        script{
            bat """
            echo Running Java Program...
            cd cloneJava
            java HelloWorld
            """
        }
    }
}
}
}
}
}
}
}

```

Output:

```

Started by user Devops
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\java_pipeline
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Tool Install)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone Repository)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] script
[Pipeline] {
[Pipeline] fileExists
[Pipeline] bat

```

```

C:\ProgramData\Jenkins\.jenkins\workspace\java_pipeline>rmdir /S /Q "C:\ProgramData\Jenkins\.jenkins\workspace\java_pipeline\cloneJava"
[Pipeline] echo
Removed existing Repo folder.
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\java_pipeline>mkdir cloneJava
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\java_pipeline>git clone https://github.com/kaverikadamm/Practical4.git
"C:\ProgramData\Jenkins\.jenkins\workspace\java_pipeline\cloneJava"
Cloning into 'C:\ProgramData\Jenkins\.jenkins\workspace\java_pipeline\cloneJava'...
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Compile Java Program)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] bat

```

```

C:\ProgramData\Jenkins\.jenkins\workspace\java_pipeline>echo Compiling Java Program
Compiling Java Program

C:\ProgramData\Jenkins\.jenkins\workspace\java_pipeline>cd cloneJava

C:\ProgramData\Jenkins\.jenkins\workspace\java_pipeline\cloneJava>javac HelloWorld.java
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Run Java Program)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\java_pipeline>echo Running Java Program...
Running Java Program...

C:\ProgramData\Jenkins\.jenkins\workspace\java_pipeline>cd cloneJava

C:\ProgramData\Jenkins\.jenkins\workspace\java_pipeline\cloneJava>java HelloWorld
Hello World!
[Pipeline] }

```

```

[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

```

2. Create a Jenkins pipeline to automate the deployment of a simple Node.js application that is managed using GitHub. The pipeline should:

- Clone the Node.js application from a GitHub repository.
- Install dependencies using npm install.
- Deploy the application to a local server.
- Automatically trigger the pipeline whenever changes are pushed to the GitHub repository.

Create a NodejsPract folder>>create **app.js**

In vscode terminal:

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS D:\MCA\SEM II\DevOps Pract> cd NodejsPract
● PS D:\MCA\SEM II\DevOps Pract\NodejsPract> npm init -y

```

app.js:

```
const http = require('http');

const server = http.createServer((req, res) => {
  res.write("Hello from Node.js Jenkins Pipeline!");
  res.end();
});

server.listen(3000, () => {
  console.log("Server running on http://localhost:3000");
});
```

Create a repository as NodejsPractical>> In gitbash:

```
kaver@KAVERI MINGW64 ~
$ cd d:

kaver@KAVERI MINGW64 /d
$ cd D:/MCA/"SEM II"/"DevOps Pract"/NodejsPract

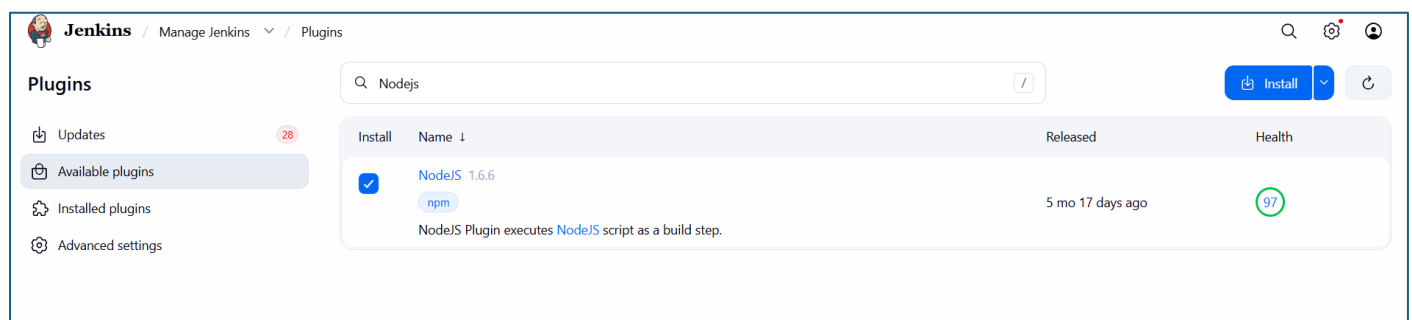
kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Pract/NodejsPract
$ git init
Initialized empty Git repository in D:/MCA/SEM II/DevOps Pract/NodejsPract/.git/
kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Pract/NodejsPract (master)
$ git add .
warning: in the working copy of 'package.json', LF will be replaced by CRLF the
next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Pract/NodejsPract (master)
$ git commit -m "Initial commit - Nodejs program"
[master (root-commit) 860cfd7] Initial commit - Nodejs program
 2 files changed, 22 insertions(+)
 create mode 100644 app.js
 create mode 100644 package.json

kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Pract/NodejsPract (master)
$ git remote add origin https://github.com/kaverikadamm/NodejsPractical.git

kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Pract/NodejsPract (master)
$ git push -u origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 594 bytes | 594.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kaverikadamm/NodejsPractical.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

Install nodejs plugin in Jenkins



The screenshot shows the Jenkins 'Manage Jenkins' interface, specifically the 'Plugins' tab. A search bar at the top contains 'Nodejs'. Below the search bar, a table lists the installed plugins. The 'NodeJS' plugin is highlighted, showing it is version 1.6.6, released 5 months and 17 days ago, and has a health status of 97. The description for the NodeJS plugin states: 'NodeJS Plugin executes NodeJS script as a build step.'

Install	Name ↓	Released	Health
☒	NodeJS 1.6.6 npm	5 mo 17 days ago	97

Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

Config File Provider

✓ Success

NodeJS

✓ Success

Loading plugin extensions

✓ Success

NodeJS

✓ Already Installed

Loading plugin extensions

✓ Success

→ [Go back to the top page](#)

(you can start using the installed plugins right away)

→ ☐ Restart Jenkins when installation is complete and no jobs are running

Jenkins pipeline 'NodejsPipeline' script:

```
pipeline {
  agent any
  environment {
    APP_NAME = "NodejsApp"
    BUILD_VERSION = "1.0"
  }
  triggers {
    githubPush()
  }
  stages {
    stage('Clone Repository') {
      steps {
        script {
          def workspace = env.WORKSPACE
          def cloneNode = "${workspace}\\cloneNode"
          if (fileExists(cloneNode)) {
            bat "rmdir /S /Q \"${cloneNode}\""
            echo "Removed existing Repo Folder"
          }
          bat "mkdir cloneNode"
          bat "git clone https://github.com/kaverikadamm/NodejsPractical.git"
          "${workspace}\\cloneNode\"
        }
      }
    }
    stage('Clean Old Deployment') {
      steps {
        script {
          def workspace = env.WORKSPACE
          def deployPath = "${workspace}\\${APP_NAME}"
          if (fileExists(deployPath)) {
            bat "rmdir /S /Q \"${deployPath}\""
            echo "Removed existing App Folder"
          }
        }
      }
    }
  }
}
```

```

    }
  }
  stage('Copy Required Files') {
    steps {
      script {
        def workspace = env.WORKSPACE
        def sourceDir = "${workspace}\\cloneNode"
        def destinationDir = "${workspace}\\${APP_NAME}"
        bat "mkdir \"${destinationDir}\""
        bat "xcopy /S /E /Y \"${sourceDir}\\*\" \"${destinationDir}\\\""
        echo "Files copied successfully."
      }
    }
  }
  stage('Install Dependencies') {
    steps {
      script {
        def workspace = env.WORKSPACE
        bat "cd /d \"${workspace}\\${APP_NAME}\" && npm install"
      }
    }
  }
  stage('Deploy Application') {
    steps {
      script {
        def workspace = env.WORKSPACE
        bat "cd /d \"${workspace}\\${APP_NAME}\" && start node app.js"
      }
    }
  }
  stage('Open Application') {
    steps {
      script {
        def appUrl = "http://localhost:3000/"
        bat "start \"\" \"${appUrl}\""
      }
    }
  }
}
post {
  success {
    echo "Application deployed successfully!"
  }
  failure {
    echo "Deployment failed!"
  }
}
}

```

Output:

Started by user Devops

[Pipeline] Start of Pipeline

[Pipeline] node

Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline

[Pipeline] {

[Pipeline] withEnv

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Clone Repository)

[Pipeline] script

[Pipeline] {

[Pipeline] fileExists

[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline>rmdir /S /Q "C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline\cloneNode"

[Pipeline] echo

Removed existing Repo Folder

[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline>mkdir cloneNode

[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline>git clone <https://github.com/kaverikadamm/NodejsPractical.git>

"C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline\cloneNode"

Cloning into 'C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline\cloneNode'...

[Pipeline] }

[Pipeline] // script

[Pipeline] }

[Pipeline] {

[Pipeline] fileExists

[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline>rmdir /S /Q "C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline\NodejsApp"

[Pipeline] echo

Removed existing App Folder

[Pipeline] }

[Pipeline] // script

[Pipeline] }

[Pipeline] // stage

[Pipeline] stage

[Pipeline] { (Copy Required Files)

[Pipeline] script

[Pipeline] {

[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline>mkdir "C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline\NodejsApp"

[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline>xcopy /S /E /Y "C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline\cloneNode**"

"C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline\NodejsApp\"

C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline\cloneNode\app.js

C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline\cloneNode\package.json

2 File(s) copied

[Pipeline] echo

Files copied successfully.


```

[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Install Dependencies)
[Pipeline] script
[Pipeline] {
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline>cd /d "C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline\NodejsApp" && npm
install

up to date, audited 1 package in 449ms

found 0 vulnerabilities
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy Application)
[Pipeline] script
[Pipeline] {
[Pipeline] bat

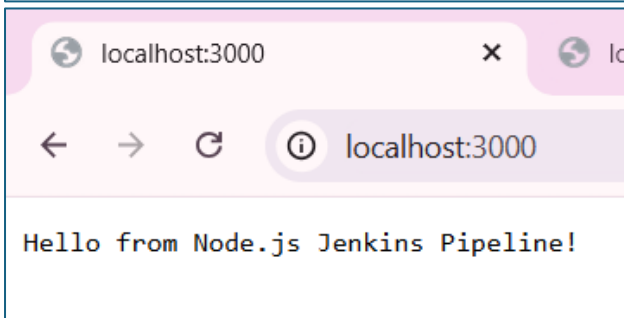
```

```

C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline>cd /d "C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline\NodejsApp" && start
node app.js
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Open Application)
[Pipeline] script
[Pipeline] {
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\NodejsPipeline>start "" "http://localhost:3000/"

```



3. Create a Jenkins pipeline to automate the deployment of a simple JSP application that is stored on the developer's machine. The pipeline should:

- a. Fetch the JSP application from the developer's local machine.
- b. Deploy the application to an Apache Tomcat server on the developer's machine.
- c. Ensure the application is accessible via a web browser.

Create a new Dynamic Web Project Click on new runtime.

New Dynamic Web Project

Dynamic Web Project

Create a standalone Java-based Web Application or add it to a new or existing Enterprise Application.

Project name:

Project location

☒ Use default location

Location:

Target runtime

Dynamic web module version

Select Apache10.1. Click on Browse and browse to tomcat version >> Finish.

Select the type of runtime environment:

type filter text

- Apache Tomcat v8.5
- Apache Tomcat v9.0
- Apache Tomcat v10.0
- Apache Tomcat v10.1**
- > Basic
- > IBM

New Server Runtime Environment

Tomcat Server

Specify the Tomcat installation directory and JRE for this runtime. The specified JRE controls the highest supported Java Facet version.

Name:

Tomcat installation directory:

apache-tomcat-10.1.12

> This PC > Windows (C:) > Program Files > Apache Software Foundation > Tomcat 10.1 >

Generate web.xml deployment descriptor. Create jsp files.

New Dynamic Web Project

Web Module

Configure web module settings.

Context root:

Content directory:

☒ Generate web.xml deployment descriptor

- StudentForm
 - Deployment Descriptor: StudentForm
 - JAX-WS Web Services
 - Java Resources
 - build
 - src
 - main
 - java
 - webapp**
 - META-INF
 - WEB-INF
 - index.jsp
 - result.jsp

To start tomcat server, open cmd as administrator and run below commands

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd "C:\Program Files\Apache Software Foundation\Tomcat 10.1\bin"
C:\Program Files\Apache Software Foundation\Tomcat 10.1\bin>startup.bat
```

Run tomcat : localhost:8080

Open Jenkins: localhost:8090

New Item

Enter an item name

JSPWebPipeline26

Select an item type

 Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and run

Script:

```
pipeline{
  agent any

  stages{
    stage('Clean old deployment'){
      steps{
        script{
          def workspace=env.WORKSPACE

          //remove existing JSPPract folder if exists
          def jspDemoPath="${workspace}\\JSPPract"
          if(fileExists(jspDemoPath)){
            bat "rmdir /S /Q \"${JSPPractPath}\""
            echo "Removed existing JSPPract Folder."
          }

          //remove any existing WAR file
          def warFile="${workspace}\\JSPPract26.war"
          if(fileExists(warFile)){
            bat "del /F /Q \"${warFile}\""
            echo "Removed existing WAR file."
          }
        }
      }
    }
    stage('Copy Required Files'){
      steps{
        script{
          def workspace=env.WORKSPACE
          def sourceDir="D:\\MCA\\SEM II\\DevOps Lab\\StudentForm\\src\\main\\webapp"
          def destinationDir="${workspace}\\JSPPract"

          //create JSPDemo folder
          bat "mkdir \"${destinationDir}\""

          //copy .jsp
          bat "xcopy /S /E /Y \"${sourceDir}\\*.jsp\" \"${destinationDir}\""

          //copy WEB-INF folder
```



```

C:\ProgramData\Jenkins\.jenkins\workspace\JSPWebPipeline26@2>xcopy /S /E /Y "D:\MCA\SEM II\DevOps Lab\StudentForm\src\main\webapp\*.jsp"
"C:\ProgramData\Jenkins\.jenkins\workspace\JSPWebPipeline26@2\JSPPract\"
D:\MCA\SEM II\DevOps Lab\StudentForm\src\main\webapp\index.jsp
D:\MCA\SEM II\DevOps Lab\StudentForm\src\main\webapp\result.jsp
2 File(s) copied
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\JSPWebPipeline26@2>xcopy /S /E /Y "D:\MCA\SEM II\DevOps Lab\StudentForm\src\main\webapp\WEB-INF"
"C:\ProgramData\Jenkins\.jenkins\workspace\JSPWebPipeline26@2\JSPPract\WEB-INF\"
D:\MCA\SEM II\DevOps Lab\StudentForm\src\main\webapp\WEB-INF\web.xml
1 File(s) copied
[Pipeline] echo
Files copied successfully!
[Pipeline] echo
"C:\ProgramData\Jenkins\.jenkins\workspace\JSPWebPipeline26@2\JSPPract"
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\JSPWebPipeline26@2>C:\PROGRA~1\Java\jdk-17\bin\jar -cvf JSPPract26.war -C
"C:\ProgramData\Jenkins\.jenkins\workspace\JSPWebPipeline26@2\JSPPract" .
added manifest
ignoring entry META-INF/
adding: WEB-INF/(in = 0) (out= 0)(stored 0%)
adding: WEB-INF/lib/(in = 0) (out= 0)(stored 0%)
adding: WEB-INF/web.xml(in = 660) (out= 264)(deflated 60%)
adding: index.jsp(in = 542) (out= 295)(deflated 45%)
adding: result.jsp(in = 576) (out= 307)(deflated 46%)
[Pipeline] }
[Pipeline] // script

```

```

[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] bat

```

```

C:\ProgramData\Jenkins\.jenkins\workspace\JSPWebPipeline26@2>curl -u admin:admin --upload-file JSPPract26.war
"http://localhost:8080/manager/text/deploy?path=/JSPPract26&update=true"

```

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current	
			Dload	Upload	Total	Spent	Left	Speed
0	0	0	0	0	0	0	0	0
100	1800	0	57 100	1743	254	7785		0
100	1800	0	57 100	1743	252	7716		0
100	1800	0	57 100	1743	250	7653		0

```

OK - Deployed application at context path [/JSPPract26]

```

```

[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy WAR to Tomcat)
[Pipeline] script
[Pipeline] {
[Pipeline] bat

```

```

C:\ProgramData\Jenkins\.jenkins\workspace\JSPWebPipeline26@2>start firefox http://localhost:8080/JSPPract26/

```

← → ↻ ⓘ localhost:8080/JSPPract26/

Student Registration Form

Name:

Age:

Course:

Student Registration Form

Name:

Age:

Course:

Submitted Details

Name: kaveri

Age: 21

Course: DevOps

[Go Back](#)

4. Create a Jenkins project that uses "Execute Windows batch command" to run a simple Node.js application stored on the developer's local machine. The project should:

- Fetch the Node.js application from the developer's machine.**
- Install dependencies using npm install.**
- Start the Node.js application using node index.js.**
- Ensure the application is running and accessible.**
- Enable automatic execution whenever changes are made to the application.**

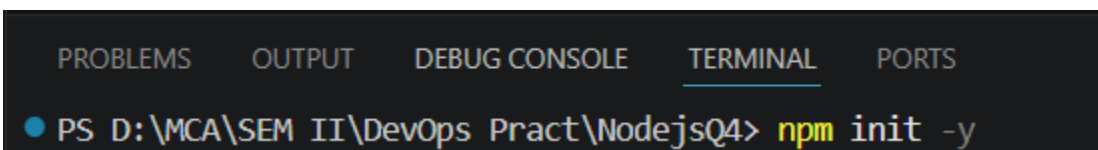
Create folder NodejsQ4

Index.js:

```
const http = require('http');

const server = http.createServer((req, res) => {
  res.write("Hello from Local Node App via Jenkins!");
  res.end();
});

server.listen(3100, () => {
  console.log("Server running at http://localhost:3100");
});
```



Create pipeline NodejsLocalPipeline

Script:

```
pipeline {
  agent any

  triggers {
    pollSCM('H/5 * * * *') // runs every 5 minutes
  }

  environment {
```

```

APP_NAME = "NodeLocalApp"
}

stages {

stage('Clean Old Deployment') {
    steps {
        script {
            def workspace = env.WORKSPACE
            //reomve existing JSPPract folder if exists
            def deployPath = "${workspace}\\${APP_NAME}"

            if (fileExists(deployPath)) {
                bat "rmdir /S /Q \"${deployPath}\""
                echo "Removed existing App Folder"
            }
        }
    }
}

stage('Fetch Local Application') {
    steps {
        script {
            def workspace = env.WORKSPACE
            def sourceDir = "D:\\MCA\\SEM II\\DevOps Pract\\NodejsQ4"
            def destinationDir = "${workspace}\\${APP_NAME}"

            bat "mkdir \"${destinationDir}\""
            bat "xcopy /S /E /Y \"${sourceDir}\\*\" \"${destinationDir}\\\""
            echo "Application fetched from local machine"
        }
    }
}

stage('Install Dependencies') {
    steps {
        script {
            def workspace = env.WORKSPACE
            bat "cd /d \"${workspace}\\${APP_NAME}\" && npm install"
        }
    }
}

stage('Start Node App') {
    steps {
        script {
            def workspace = env.WORKSPACE

            bat "cd /d \"${workspace}\\${APP_NAME}\" && start node index.js"
        }
    }
}

stage('Open Application') {

```

```

    steps {
        script {
            def appUrl = "http://localhost:3100/"
            bat "start \"\" \"${appUrl}\""
        }
    }
}
stage('Deploy Application'){
    steps {
        script {
            bat "timeout /t 5"
            bat "curl http://localhost:3100"
        }
    }
}
}
post {
    success {
        echo "Application deployed successfully!"
    }
    failure {
        echo "Deployment failed!"
    }
}
}
}

```

Output:

```

Started by user Devops
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\NodejsLocalPipeline
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clean Old Deployment)
[Pipeline] script
[Pipeline] {
[Pipeline] fileExists
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\NodejsLocalPipeline>rmdir /S /Q
"C:\ProgramData\Jenkins\.jenkins\workspace\NodejsLocalPipeline\NodeLocalApp"
[Pipeline] echo
Removed existing App Folder
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Fetch Local Application)
[Pipeline] script
[Pipeline] {
[Pipeline] bat

```



```
C:\ProgramData\Jenkins\.jenkins\workspace\NodejsLocalPipeline>mkdir
"C:\ProgramData\Jenkins\.jenkins\workspace\NodejsLocalPipeline\NodeLocalApp"
[Pipeline] bat
```

```
C:\ProgramData\Jenkins\.jenkins\workspace\NodejsLocalPipeline>xcopy /S /E /Y "D:\MCA\SEM II\DevOps Pract\NodejsQ4\*"
"C:\ProgramData\Jenkins\.jenkins\workspace\NodejsLocalPipeline\NodeLocalApp\"
D:\MCA\SEM II\DevOps Pract\NodejsQ4\index.js
D:\MCA\SEM II\DevOps Pract\NodejsQ4\package.json
2 File(s) copied
[Pipeline] echo
Application fetched from local machine
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Install Dependencies)
[Pipeline] script
[Pipeline] {
[Pipeline] bat
```

```
C:\ProgramData\Jenkins\.jenkins\workspace\NodejsLocalPipeline>cd /d
"C:\ProgramData\Jenkins\.jenkins\workspace\NodejsLocalPipeline\NodeLocalApp"  && npm install

up to date, audited 1 package in 622ms

found 0 vulnerabilities
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Start Node App)
[Pipeline] script
[Pipeline] {
[Pipeline] bat
```

```
C:\ProgramData\Jenkins\.jenkins\workspace\NodejsLocalPipeline>cd /d
"C:\ProgramData\Jenkins\.jenkins\workspace\NodejsLocalPipeline\NodeLocalApp"  && start node index.js
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Open Application)
[Pipeline] script
[Pipeline] {
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\NodejsLocalPipeline>start "" "http://localhost:3100/"
```

← → ↻ ⓘ localhost:3100

Hello from Local Node App via Jenkins!